Confidence Intervals & Regions $(1-\alpha)$

- Single Normal Distribution

σ^2 known, μ: $[\bar{X}-\frac{\sigma}{\sqrt{n}}z_{\alpha/2}, \bar{X}+\frac{\sigma}{\sqrt{n}}z_{\alpha/2}]$

σ^2 unknown, μ: $\sigma \to S$

μ known, σ^2: $[\frac{nS_\mu^2}{\chi^2_{n,\alpha/2}}, \frac{nS_\mu^2}{\chi^2_{n,1-\alpha/2}}]$, $S_\mu^2=\frac{1}{n}\sum(X_i-\mu)^2$

μ knownn, σ^2: $S_\mu^2 \to S^2$, $n \to n-1$

- X_i from $U(\theta-\frac{1}{2},\theta+\frac{1}{2})$ θ: $[\frac{X_{(1)}+X_{(n)}}{2} \pm \frac{1-\sqrt[n]{\alpha}}{2}]$

$X_{(1)}=n(\theta+\frac{1}{2}-x)^{n-1}I_{|x-\theta|<\frac{1}{2}}$ $X_{(n)}$: $f(x)=n(x-\theta+\frac{1}{2})^{n-1}I_{|x-\theta|<\frac{1}{2}}$

$Z=\frac{X_{(1)}-\theta+X_{(n)}-\theta}{2}$ $f(z)=n(1-2|z|)^{n-1}, -\frac{1}{2}\le z\le\frac{1}{2}$

- X_i from $f(x,\lambda)=\lambda e^{-\lambda x}I_{x>0}$ λ: $[\frac{\chi^2_{2n,1-\alpha/2}}{2n\bar{X}}, \frac{\chi^2_{2n,\alpha/2}}{2n\bar{X}}]$

$Z=2\lambda n\bar{X}=2\lambda\sum X_i \sim \chi^2_{2n}$

- $X_m \sim N(\mu_1,\sigma_1^2)$ $Y_n \sim N(\mu_2,\sigma_2^2)$

$\sigma_1^2=\sigma_2^2$ unknown $\mu_1-\mu_2$: $[(\bar{X}-\bar{Y}) \pm t_{m+n-2}(\frac{\alpha}{2})\sqrt{\frac{(m-1)S_X^2+(n-1)S_Y^2}{m+n-2}(\frac{1}{m}+\frac{1}{n})}]$

$\frac{\sigma_1^2}{\sigma_2^2}$: $[\frac{S_X^2}{S_Y^2}F_{n-1,m-1;1-\alpha/2}; \frac{S_X^2}{S_Y^2}F_{n-1,m-1;\alpha/2}]$ $Z=\frac{S_Y^2/\sigma_2^2}{S_X^2/\sigma_1^2}\sim F_{n-1,m-1}$

CLT: $X_i \sim B(1,\theta)$ $Z=\frac{\sqrt{n}(\bar{X}-\theta)}{\sqrt{\bar{X}(1-\bar{X})}}$ θ: $[\bar{X}\pm z_{\alpha/2}\sqrt{\frac{\bar{X}(1-\bar{X})}{n}}]$

$X_i \sim P(\theta)$ $Z=\frac{\sqrt{n}(\bar{X}-\theta)}{\sqrt{\bar{X}}}$ θ: $[\bar{X}\pm z_{\alpha/2}\sqrt{\frac{\bar{X}}{n}}]$

$X_i \sim \mathcal{E}(\frac{1}{\theta})$ $Z=\frac{\sqrt{n}(\bar{X}-\theta)}{\bar{X}}$ θ: $[\bar{X}\pm z_{\alpha/2}\frac{\bar{X}}{\sqrt{n}}]$

$\sigma_1^2 \neq \sigma_2^2$ $Z=\frac{\bar{X}-\bar{Y}-(\mu_1-\mu_2)}{\sqrt{S_X^2/m+S_Y^2/n}}$ $\mu_1-\mu_2$: $[\bar{X}-\bar{Y}\pm z_{\alpha/2}\sqrt{\frac{S_X^2}{m}+\frac{S_Y^2}{n}}]$

$\hookrightarrow T_* \sim T_r$ (r nearest $(\frac{S_X^2}{m}+\frac{S_Y^2}{n})^2/[\frac{S_X^4}{m^2(m-1)}+\frac{S_Y^4}{n^2(n-1)}]$)

Confidence Region: $X_i \sim N(\mu,\sigma^2)$ (μ,σ^2) $\to \begin{cases} P[|\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}|\le c]=\sqrt{1-\alpha} \\ P[a\le\frac{(n-1)S^2}{\sigma^2}\le b]=\sqrt{1-\alpha}\end{cases}$

$P[(\mu-\bar{X})^2\le\frac{c^2}{n}\sigma^2]\cdot P[\frac{(n-1)S^2}{b}\le\sigma^2\le\frac{(n-1)S^2}{a}]=1-\alpha$

$\Rightarrow \gamma=\frac{1-\sqrt{1-\alpha}}{2}$, $c=z_\gamma^2$, $b=\chi^2_{n-1,\gamma}$ $a=\chi^2_{n-1,1-\gamma}$

- $X_m \sim N(\mu_1,\sigma_1^2)$ $Y_n \sim N(\mu_2,\sigma_2^2)$ (μ_1,μ_2)

$\begin{cases} \text{independent} & [\bar{X}\pm t_{m-1,\gamma}\frac{S_X}{\sqrt{m}}]\times[\bar{Y}\pm t_{n-1,\gamma}\frac{S_Y}{\sqrt{n}}] \\ \text{dependent} & [\bar{X}\pm t_{n-1,\frac{\alpha}{4}}\frac{S_X}{\sqrt{n}}]\times[\ ]\ (m=n)\end{cases}$

– Hypothesis Testing

$\Pi_\varphi(\theta)$ 可能需要分类

$\alpha=\max\alpha(\theta), \theta\in\Theta_0$, $\alpha(\theta)=\begin{cases}\Pi_\varphi(\theta), & \theta\in\Theta_0 \\ 0, & \theta\in\Theta_1\end{cases}$ $\beta(\theta)=\begin{cases}0, & \theta\in\Theta_0 \\ 1-\Pi_\varphi(\theta), & \theta\in\Theta_1\end{cases}$

p-value $p=P(|T(X)|\ge|t(x)|\,|H_0)$ – two side test

$p<\alpha \Leftrightarrow t(x)$ in rejection region D

- Single Normal Population $X_i \sim N(\mu,\sigma^2)$

H_0: $\mu\le\mu_0$ → 先取$\mu=\mu_0$，证明H_0下 $\Pi(\mu)\le\alpha$，求$\Pi(\theta)$拆成正态形式

μ	H_0	H_1	Test Statistic and its Distribution	Rejection Region
σ^2 known	$\mu=\mu_0$	$\mu\neq\mu_0$	$Z=\sqrt{n}(\bar{X}-\mu_0)/\sigma$ $Z\mid\mu=\mu_0\sim N(0,1)$	$\|Z\|>z_{\alpha/2}$
	$\mu\le\mu_0$	$\mu>\mu_0$		$Z>z_\alpha$
	$\mu\ge\mu_0$	$\mu<\mu_0$		$Z<-z_\alpha$
σ^2 unknown	$\mu=\mu_0$	$\mu\neq\mu_0$	$T=\sqrt{n}(\bar{X}-\mu_0)/S$ $T\mid\mu=\mu_0\sim t_{n-1}$	$\|T\|>t_{n-1,\alpha/2}$
	$\mu\le\mu_0$	$\mu>\mu_0$		$T>t_{n-1,\alpha}$
	$\mu\ge\mu_0$	$\mu<\mu_0$		$T<-t_{n-1,\alpha}$

H_0: $\mu=\mu_0 \leftrightarrow H_1$: $\mu=\mu_1(\mu_1>\mu_0)$ one-side test better

σ^2	H_0	H_1	Test Statistic and its Distribution	Rejection Region
μ known	$\sigma^2=\sigma_0^2$	$\sigma^2\neq\sigma_0^2$	$\chi_\mu^2=nS_\mu^2/\sigma_0^2$ $\chi_\mu^2\mid\sigma^2=\sigma_0^2\sim\chi_n^2$	$\frac{nS_\mu^2}{\sigma_0^2}<\chi^2_{n,1-\alpha/2}$ or $\frac{nS_\mu^2}{\sigma_0^2}>\chi^2_{n,\alpha/2}$
	$\sigma^2\le\sigma_0^2$	$\sigma^2>\sigma_0^2$		$\frac{nS_\mu^2}{\sigma_0^2}>\chi^2_{n,\alpha}$
	$\sigma^2\ge\sigma_0^2$	$\sigma^2<\sigma_0^2$		$\frac{nS_\mu^2}{\sigma_0^2}<\chi^2_{n,1-\alpha}$
μ unknown	$\sigma^2=\sigma_0^2$	$\sigma^2\neq\sigma_0^2$	$\chi^2=\frac{(n-1)S^2}{\sigma_0^2}$ $\chi^2\mid\sigma^2=\sigma_0^2\sim\chi^2_{n-1}$	$\frac{(n-1)S^2}{\sigma_0^2}<\chi^2_{n-1,1-\frac{\alpha}{2}}$ or $\frac{(n-1)S^2}{\sigma_0^2}>\chi^2_{n-1,\frac{\alpha}{2}}$
	$\sigma^2\le\sigma_0^2$	$\sigma^2>\sigma_0^2$		$\frac{(n-1)S^2}{\sigma_0^2}>\chi^2_{n-1,\alpha}$
	$\sigma^2\ge\sigma_0^2$	$\sigma^2<\sigma_0^2$		$\frac{(n-1)S^2}{\sigma_0^2}<\chi^2_{n-1,1-\alpha}$

- Test for $B(1,\theta)$: randomized test : $T(X)=\sum X_i \sim B(n,\theta)$

H_0: $\theta=\theta_0 \leftrightarrow H_1$: $\theta\neq\theta_0$ $\varphi(X)=\begin{cases}1 & \\ r_1 & T(X)=c_1 \\ r_2 & T(X)=c_2 \\ 0 & \end{cases}$ $P_{\theta_0}(T(X)<c_1)+r_1P_{\theta_0}(T(X)=c_1)=\frac{\alpha}{2}$

- Two Normal Populations : $X_m \sim N(\mu_1,\sigma_1^2)$ $Y_n \sim N(\mu_2,\sigma_2^2)$

$\mu_1-\mu_2$	H_0	H_1	Test Statistic and its Distribution	Rejection Region
σ_1^2,σ_2^2 known	$\mu_1-\mu_2=\mu_0$	$\mu_1-\mu_2\neq\mu_0$	$U=\frac{\bar{X}-\bar{Y}-\mu_0}{\sqrt{\sigma_1^2/m+\sigma_2^2/n}}$ $U\mid\mu_1-\mu_2=\mu_0\sim N(0,1)$	$\|U\|>z_{\alpha/2}$
	$\mu_1-\mu_2\le\mu_0$	$\mu_1-\mu_2>\mu_0$		$U>z_\alpha$
	$\mu_1-\mu_2\ge\mu_0$	$\mu_1-\mu_2<\mu_0$		$U<-z_\alpha$
$\sigma_1^2=\sigma_2^2=\sigma^2$ unknown	$\mu_1-\mu_2=\mu_0$	$\mu_1-\mu_2\neq\mu_0$	$T=\frac{\bar{X}-\bar{Y}-\mu_0}{\sqrt{\frac{(m-1)S_X^2+(n-1)S_Y^2}{m+n-2}(\frac{1}{m}+\frac{1}{n})}}$ $T\mid\mu_1-\mu_2=\mu_0\sim t_{m+n-2}$	$\|T\|>t_{m+n-2,\alpha/2}$
	$\mu_1-\mu_2\le\mu_0$	$\mu_1-\mu_2>\mu_0$		$T>t_{m+n-2,\alpha}$
	$\mu_1-\mu_2\ge\mu_0$	$\mu_1-\mu_2<\mu_0$		$T<-t_{m+n-2,\alpha}$

$\sigma_1^2\neq\sigma_2^2$ unknown: $T_*=\frac{\bar{X}-\bar{Y}-\mu_0}{\sqrt{S_X^2/m+S_Y^2/n}}$ 同上近似分布 T_r

Paired Comparison $(m=n)$ $Z_i=Y_i-X_i$ H_0: $\mu=0$

σ_1^2/σ_2^2	H_0	H_1	Test Statistic and its Distribution	Rejection Region
μ_1,μ_2 known	$\sigma_2^2=\sigma_1^2$	$\sigma_2^2\neq\sigma_1^2$	$F_*=\frac{S_{Y*}^2}{S_{X*}^2}$ $F_*\mid_{\sigma_2^2=\sigma_1^2}\sim F_{n,m}$	$F_*<F_{n,m,1-\alpha/2}$ or $F_*>F_{n,m,\alpha/2}$
	$\sigma_2^2\le\sigma_1^2$	$\sigma_2^2>\sigma_1^2$		$F_*>F_{n,m,\alpha}$
	$\sigma_2^2\ge\sigma_1^2$	$\sigma_2^2<\sigma_1^2$		$F_*<F_{n,m,1-\alpha}$
μ_1,μ_2 unknown	$\sigma_2^2=\sigma_1^2$	$\sigma_2^2\neq\sigma_1^2$	$F=\frac{S_Y^2}{S_X^2}$ $F\mid_{\sigma_2^2=\sigma_1^2}\sim F_{n-1,m-1}$	$F<F_{n-1,m-1,1-\frac{\alpha}{2}}$ or $F>F_{n-1,m-1,\frac{\alpha}{2}}$
	$\sigma_2^2\le\sigma_1^2$	$\sigma_2^2>\sigma_1^2$		$F>F_{n-1,m-1,\alpha}$
	$\sigma_2^2\ge\sigma_1^2$	$\sigma_2^2<\sigma_1^2$		$F<F_{n-1,m-1,1-\alpha}$

CLT: $U^*=\frac{\bar{X}-\bar{Y}-\mu_0}{\sqrt{S_X^2/m+S_Y^2/n}}\xrightarrow{d}N(0,1)$ $F_{m,n,1-\alpha/2}=1/F_{n,m,\alpha/2}$

$B(1,\theta)$ $U=\frac{\sqrt{n}(\bar{X}-\theta_0)}{\sqrt{\theta_0(1-\theta_0)}}\xrightarrow{d}N(0,1)$

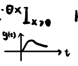
$P(\theta)$ $U=\frac{\sqrt{n}(\bar{X}-\theta_0)}{\sqrt{\theta_0}}\xrightarrow{d}N(0,1)$

LR test

$\lambda(x)=\frac{L(\hat\theta_{MLE,H_0})}{L(\hat\theta_{MLE})}$ $\varphi(x)=\begin{cases}1, & \lambda<\lambda_0 \\ r, & \lambda=\lambda_0 \\ 0, & \lambda>\lambda_0\end{cases}$

- $X_i \sim f(x,\theta)=\theta e^{-\theta x}I_{x>0}$ H_0: $\theta=\theta_0$ let $t=\sum^n x_i$

$\lambda=(\frac{e\theta_0}{n})^n t^n e^{-\theta_0 t}$



$t<c_1$ or $t>c_2$

χ^2_{2n} 与区间估计对偶

- Single Normal Population $X_i \sim N(\mu,\sigma^2)$ $\hat\mu=\bar{X}$ $\hat\sigma^2=\frac{1}{n}\sum(X_i-\hat\mu)^2$

(i) σ known H_0: $\mu=\mu_0$ → $-2\log\lambda=[\frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma}]^2$

(ii) σ unknown H_0: $\mu=\mu_0$ $L(\hat\omega)=(2\pi\hat\sigma_\omega^2)^{-\frac{n}{2}}e^{-\frac{n}{2}}$ $L(\hat\Omega)=(2\pi\hat\sigma_\Omega^2)^{-\frac{n}{2}}e^{-\frac{n}{2}}$

$\lambda^{\frac{2}{n}}=\frac{\sum(X_i-\bar{X})^2}{\sum(X_i-\mu_0)^2}$ let $t=\sqrt{n}(\bar{X}-\mu_0)/\sqrt{\frac{1}{n-1}\sum(X_i-\bar{X})^2}$

$\lambda^{\frac{2}{n}}=1/[1+\frac{t^2}{n-1}]$ T-test

H_0: $\mu\le\mu_0$ 分类 $L(\hat\omega)=\begin{cases}L(\hat\Omega), & \bar{X}\le\mu_0 \\ (2\pi\hat\sigma_\omega)^{-\frac{n}{2}}e^{-\frac{n}{2}}, & \bar{X}>\mu_0\end{cases}$ $\Rightarrow\lambda=\begin{cases}1, & \bar{X}\le\mu_0 \\ [1+\frac{t^2}{n-1}]^{-\frac{n}{2}}, & \bar{X}>\mu_0\end{cases}$

(iii) μ known, H_0: $\sigma=\sigma_0$ $\Rightarrow\lambda=e^{\frac{n}{2}}u^{\frac{n}{2}}\exp(-\frac{nu}{2})$, $u=\frac{1}{n}\sum_{i=1}^n(\frac{X_i-\mu}{\sigma_0})^2$ | χ^2-test

(iv) μ unknown, H_0: $\sigma=\sigma_0$ $\Rightarrow\lambda=e^{\frac{n}{2}}(\frac{\hat\sigma_\Omega^2}{\sigma_0^2})^{\frac{n}{2}}\exp(-\frac{nu}{2})$

- Two Normal Populations $X_m \sim N(\mu_1,\sigma_1^2)$ $Y_n \sim N(\mu_2,\sigma_2^2)$

(i) $\sigma_1=\sigma_2$ unknown H_0: $\mu_1-\mu_2=0$ → $L(\hat\Omega)=(2\pi\hat\sigma_\Omega^2)^{-\frac{m+n}{2}}\exp(-\frac{m+n}{2})$, $\hat\sigma_\Omega^2=\frac{1}{m+n}[\sum(X_i-\bar{X})^2+\sum(Y_i-\bar{Y})^2]$

$\lambda=(\frac{\hat\sigma_\Omega^2}{\hat\sigma_\omega^2})^{\frac{m+n}{2}}$ $\hat\mu_\omega=\frac{m\bar{X}+n\bar{Y}}{m+n}$ $\hat\sigma_\omega^2=\frac{1}{m+n}[\sum(X_i-\hat\mu)^2+\sum(Y_i-\hat\mu)^2]=\hat\sigma_\Omega^2+\frac{mn(\bar{X}-\bar{Y})^2}{(m+n)^2}$

$\Rightarrow\lambda^{\frac{2}{m+n}}=(1+\frac{t^2}{m+n-2})^{-1}$, $t=\sqrt{\frac{mn}{m+n}}(\bar{X}-\bar{Y})/\sqrt{\frac{\sum(X_i-\bar{X})^2+\sum(Y_i-\bar{Y})^2}{m+n-2}}\sim t_{m+n-2}$

(ii) μ_1,μ_2 unknown H_0: $\sigma_1=\sigma_2$ → $L(\hat\Omega)=(2\pi)^{-\frac{m+n}{2}}(\hat\sigma_{1,\Omega}^2)^{-\frac{m}{2}}(\hat\sigma_{2,\Omega}^2)^{-\frac{n}{2}}\exp(-\frac{m+n}{2})$

$\hat\sigma_{1,\Omega}^2=\frac{1}{m}\sum^m(X_i-\bar{X})^2$, $\hat\sigma_{2,\Omega}^2=\frac{1}{n}\sum^n(Y_i-\bar{Y})^2$

$L(\hat\omega)=(2\pi)^{-\frac{m+n}{2}}(\hat\sigma_\omega^2)^{-\frac{m+n}{2}}\exp(-\frac{m+n}{2})$ $\hat\sigma_\omega^2=\frac{1}{m+n}[\sum(X_i-\bar{X})^2+\sum(Y_i-\bar{Y})^2]$

$\lambda=\frac{(m+n)^{\frac{m+n}{2}}}{m^{m/2}n^{n/2}}\cdot\frac{(\frac{m-1}{n-1}u)^{m/2}}{(1+\frac{m-1}{n-1}u)^{\frac{m+n}{2}}}$, $a=\sum(X_i-\bar{X})^2$, $b=\sum(Y_i-\bar{Y})^2$, $u=\frac{a}{m-1}/\frac{b}{n-1}\sim F_{m-1,n-1}$

- θ K维, Θ_0 S维, H_0: $\theta\in\Theta_0 \leftrightarrow H_1$: $\theta\in\Theta_1$, under H_0, $n\to\infty$

$-2\log\lambda(X)\xrightarrow{d}\chi^2_{K-S}$ 不同算 $\Pi_\varphi(\theta)$

- $X_i \sim N(\mu_i,\sigma_i^2), i=1,\dots m$ X_i's independent H_0: $\sigma_1^2=\sigma_2^2=\dots\sigma_m^2$

$L(\theta,X)=\prod_{i=1}^m[(2\pi\sigma_i^2)^{-\frac{n_i}{2}}\exp\{-\frac{1}{2\sigma_i^2}\sum_{j=1}^{n_i}(X_{ij}-\mu_i)^2\}]$

$\bar{X}_i=\frac{1}{n_i}\sum_{j=1}^{n_i}X_{ij}$ $S_i^2=\frac{1}{n_i}\sum_{j=1}^{n_i}(X_{ij}-\bar{X}_i)^2$, $S^2=\frac{1}{n}\sum_{i=1}^m n_iS_i^2$, $n=\sum_{i=1}^m n_i$

$L(\hat\Omega)=(2\pi e)^{-\frac{n}{2}}\prod_{i=1}^m S_i^{-n_i}$ $L(\hat\omega)=(2\pi e)^{-\frac{n}{2}}S^{-n}$

$-2\log\lambda(X)=n\log S^2-\sum_{i=1}^m n_i\log S_i^2\xrightarrow{d}\chi^2_{m-1}$

UMP test : θ: 1d

no UMP for 2-side test

NP Lemma: X from $f(x,\theta)$ H_0: $\theta=\theta_0 \leftrightarrow H_1$: $\theta=\theta_1$

$\varphi(x)=\begin{cases}1, & f(x,\theta_1)>cf(x,\theta_0) \\ r, & = \\ 0, & <\end{cases}$ $E_{\theta_0}\varphi(X)=\alpha$ → LR test

- X from $f(x,\theta)=\theta x^{\theta-1}I_{0<x<1}, \theta>1$ H_0: $\theta=\theta_0 \leftrightarrow H_1$: $\theta=\theta_1$

$\Rightarrow c=\begin{cases}(1-\alpha)^{1/\theta_0}, & \theta_1>\theta_0 \\ \alpha^{1/\theta_0}, & \theta_1<\theta_0\end{cases}$

- if $\varphi(x)$ not depend on θ_1, H_0: $\theta=\theta_0 \leftrightarrow H_1$: $\theta>\theta_0$ also UMP (H_1: $\theta=\theta_1(\theta_1>\theta_0)$)

↳ $\varphi(x)$ must be a function of sufficient statistic

- H_0: $\theta\in\Theta_0 \leftrightarrow H_1$: $\theta\in\Theta_1$ 在以上基础，验证 $E_\theta\varphi(x)\le\alpha$, $\theta\in\Theta_0$ 即可

在指数族中 $f(x,\theta)=C(\theta)e^{Q(\theta)T(x)}h(x)$, Q严格单增, $V(x)=\sum^n T(X_i)$

H_0: $\theta\le\theta_0 \leftrightarrow H_1$: $\theta>\theta_0$ UMP $\varphi(x)=\begin{cases}1, & V(x)>c \\ r, & = \\ 0, & <\end{cases}$

H_0: $\theta\le\theta_1$ or $\theta\ge\theta_2 \leftrightarrow H_1$: $\theta_1<\theta<\theta_2$ UMP $\varphi(x)=\begin{cases}1, & c_1<V(x)<c_2 \\ r_1, & V(x)=c_1 \\ r_2, & V(x)=c_2 \\ 0, & \text{else}\end{cases}$

- $X \sim B(1,\theta)$ $f(x,\theta)=(1-\theta)\exp\{\log(\frac{\theta}{1-\theta})\cdot x\}$

$X \sim N(\theta,\sigma^2)$ $f(x,\theta)=(2\pi\sigma^2)^{-\frac{1}{2}}e^{-\frac{\theta^2}{2\sigma^2}}e^{\frac{\theta}{\sigma^2}x}e^{-\frac{x^2}{2\sigma^2}}$, $V(x)\sim N(n\theta_i,n\sigma^2), i=1,2$

Linear Regression Model

$Y=X\beta+\epsilon$ ϵ_i i.i.d $\sim N(0,\sigma^2)$

OLS: $\hat\beta=(X^TX)^{-1}X^TY$ $\hat\sigma^2=\frac{RSS}{n-p-1}$ (MLE: $\frac{1}{n}RSS$) $RSS=\|Y-X\hat\beta\|_2^2$

$E\hat\beta=\beta$ $\hat\beta\sim N(\beta,\sigma^2(X^TX)^{-1})$ $\hat\beta_j\sim N(\beta_j,\sigma^2[(X^TX)^{-1}]_{j+1,j+1})$

$\frac{(n-p-1)\hat\sigma^2}{\sigma^2}\sim\chi^2_{n-p-1}$ $\hat\beta$ is independent of $\hat\sigma^2$

CI for σ^2 $[\frac{(n-p-1)\hat\sigma^2}{\chi^2_{n-p-1,\alpha/2}}, \frac{(n-p-1)\hat\sigma^2}{\chi^2_{n-p-1,1-\alpha/2}}]$

(β_1,β_2): $[\hat\beta_1\pm\hat\sigma[(X^TX)^{-1}]^{1/2}_{2,2}t_{n-p-1,\frac{\alpha}{4}}]\times[\hat\beta_2\pm\hat\sigma[(X^TX)^{-1}]^{\frac{1}{2}}_{3,3}t_{n-p-1,\frac{\alpha}{4}}]$

H_{0j}: $\beta_j=0 \leftrightarrow H_1$: $\beta_j\neq0$, $j=0,1,\dots p$ $\varphi(X,Y)=\begin{cases}1, & |\frac{\hat\beta_j}{\hat\sigma[(X^TX)^{-1}]^{1/2}_{j+1,j+1}}|>t_{n-1-p,\alpha/2} \\ 0, & \text{else}\end{cases}$

LR-test $\hat\beta_{MLE,0}=(X_{-j}^TX_{-j})^{-1}X_{-j}^TY$ $\hat\sigma^2_{MLE,0}=\frac{1}{n}\|Y-X_{-j}\hat\beta_{MLE,0}\|_2^2$

$\lambda(x,y)=(\frac{\hat\sigma^2_{MLE}}{\hat\sigma^2_{MLE,0}})^{\frac{n}{2}}$ $-2\log\lambda(x,y)\xrightarrow{d}\chi_1^2$

H_0: $\beta_1=\beta_2=\dots=\beta_p=0 \leftrightarrow H_1$

LR-test $\lambda(x,y)=(\frac{\hat\sigma^2_{MLE}}{\hat\sigma^2})^{\frac{n}{2}}$, $\hat\sigma^2=\frac{1}{n}\|Y-\bar{y}\|_2^2$, $-2\log\lambda\xrightarrow{d}\chi_p^2$

精确: $Z_1=\frac{\|X\hat\beta_{MLE}-\bar{y}\|_2^2}{\sigma^2}\sim\chi_p^2$, $Z_2=\frac{\|Y-X\hat\beta_{MLE}\|_2^2}{\sigma^2}\sim\chi^2_{n-p-1}$. $F=\frac{Z_1/p}{Z_2/(n-p-1)}$

$\Rightarrow\lambda^{\frac{2}{n}}=1/(1+\frac{pF}{n-p-1})$

Two-way ANOVA:

$Y_{ij}=\mu+\alpha_i+\beta_j+e_{ij}$ $\sum_{i=1}^I\alpha_i=0$ $\sum_{j=1}^J\beta_j=0$ r.v. $e_{ij}\sim N(0,\sigma^2)$

$Y_{ij}\sim N(\mu+\alpha_i+\beta_j,\sigma^2)$ independent

$H_{0,A}$: $\alpha_1=\dots=\alpha_I=0$, $H_{0,B}$: $\beta_1=\dots=\beta_J=0$

Source of Variance	Sums of Squares	Degrees of Freedom	Mean Squares
Rows	$SS_A=J\sum_{i=1}^I\hat\alpha_i^2=J\sum_{i=1}^I(Y_{i.}-Y_{..})^2$ $=J\sum_i Y_{i.}^2-IJY_{..}^2$	$I-1$	$MS_A=\frac{SS_A}{I-1}$
Columns	$SS_B=I\sum_{j=1}^J\hat\beta_j^2=I\sum_{j=1}^J(Y_{.j}-Y_{..})^2$	$J-1$	$MS_B=\frac{SS_B}{J-1}$
Residual	independent $SS_e=\sum_{i=1}^I\sum_{j=1}^J(Y_{ij}-Y_{i.}-Y_{.j}+Y_{..})^2$	$(I-1)\times(J-1)$	$MS_e=\frac{SS_e}{(I-1)(J-1)}$ ↓
Total	$SS_T=\sum_{i=1}^I\sum_{j=1}^J(Y_{ij}-Y_{..})^2=\sum_{i=1}^I\sum_{j=1}^J y_{ij}^2-IJy_{..}^2$	$IJ-1$	– unbiased $\hat\sigma^2$

$SS_T=SS_e+SS_A+SS_B$ $SS_e/\sigma^2\sim\chi^2_{(I-1)(J-1)}$

$D_A=\{y: \frac{MSA}{MSe}>F_{I-1,(I-1)(J-1),\alpha}\}$

Non-Parametric Hypothesis Testing

- Sign test $S_i=\begin{cases}1, & Z_i>0 \\ -1, & <0 \\ 0, & =0\end{cases}$ suppose Z_i's is symmetric about means under H_0

$n_0=n_++n_-$, $n_+\sim B(n_0,\theta)$, H_0: $\mu=0 \Rightarrow H_0$: $\theta=0.5 \leftrightarrow H_1$: $\theta\neq0.5$

$D=\{\vec{X}: n_+\ge c$ or $n_+\le d\}$ $\sum_{i=c}^{n_0}C_{n_0}^i(\frac{1}{2})^{n_0}\le\frac{\alpha}{2}$, $d=n_0-c$ $p=\sum_{i=0}^{x_0}C_{n_0}^i(\frac{1}{2})^{n_0}+\sum_{i=n_0-x_0}^{n_0}C_{n_0}^i(\frac{1}{2})^{n_0}$

By CLT, $U=\frac{X-\frac{1}{2}n_0}{\sqrt{n_0/4}}=\frac{2X-n_0}{\sqrt{n_0}}\xrightarrow[n_0\to\infty]{d}N(0,1)$ $D=\{X: |U|>z_{\alpha/2}\}$, $p=1-\Phi(|u|)+\Phi(-|u|)$

检验中位数无需对称分布条件

- Signed Rank Sum Test: rank of $|Z_i|$, 同秩取均 eg 4.5 → 4.5×2

$W^+=\sum_{i=1}^{n_0}R_iI_{S_i>0}$ $(W=\sum_{i=1}^{n_0}S_iR_i)$, $EW^+=\frac{n_0(n_0+1)}{4}$ $\text{var}W^+=\frac{n_0(n_0+1)(2n_0+1)}{24}$

$D=\{W^+\ge c$ or $W^+\le d\}$, $d=\frac{n_0(n_0+1)}{2}-c$ CLT $U=\frac{W^+-EW^+}{\sqrt{\text{var}W^+}}\xrightarrow{d}N(0,1)$

- Two-sample Rank Test: 假设 X, Y 为同分布的平移 θ

H_0: $\theta=0 \leftrightarrow H_1$: $\theta\neq0$ $R_1, R_2\dots R_n$ 是 $Y_1,\dots Y_n$ 在 $X_1\dots X_m, Y_1\dots Y_n$ 的秩

$W=R_1+R_2+\dots+R_n$ $P(R_1=r_1,\dots R_n=r_n)=\begin{cases}\frac{1}{N(N-1)\dots(N-n+1)}, & r_1\dots r_n\le N=m+n \\ 0 & \end{cases}$

$EW=\frac{n(N+1)}{2}$ CLT $W^*=\frac{W-EW}{\sqrt{mn(N+1)/12}}\xrightarrow{d}N(0,1)$

- Goodness-of-fit Test

H_0: $X\sim F\begin{pmatrix}a_1, & a_2 & \dots & a_r \\ p_1, & p_2 & \dots & p_r\end{pmatrix}$ Discrete → 当未知参数为s r-1-s

$K_n=\sum_{i=1}^r\frac{(v_i-np_i)^2}{np_i}\xrightarrow{d}\chi^2_{r-1}$ $D=\{K_n>\chi^2_{r-1,\alpha}\}$ $p(k_0)\doteq P(\chi^2_{r-1}\ge k_0)$

for continuous 分组 $p_j=F(a_j)-F(a_{j-1})$ 变为离散 $f\ge5$

- Contingency Table Independence & Homogeneity

Independence:

H_0: X and Y are independent

$K_n=n(\sum_{i=1}^r\sum_{j=1}^s\frac{n_{ij}^2}{n_{i.}n_{.j}}-1)$

D: $\{K_n>\chi^2_{(r-1)(s-1),\alpha}\}$

$p(k_0)=P(K_n\ge k_0|H_0)\doteq P(\chi^2_{(r-1)(s-1)}\ge k_0)$

X \ Y	1	⋯	j	⋯	s	Σ
1	n_{11}	⋯	n_{1j}	⋯	n_{1s}	$n_{1\cdot}$
⋮	⋮		⋮		⋮	⋮
i	n_{i1}	⋯	n_{ij}	⋯	n_{is}	$n_{i\cdot}$
⋮	⋮		⋮		⋮	⋮
r	n_{r1}	⋯	n_{rj}	⋯	n_{rs}	$n_{r\cdot}$
Σ	$n_{\cdot1}$	⋯	$n_{\cdot j}$	⋯	$n_{\cdot s}$	n

Homogeneity: $n_{i.}$ 给定，检验 $P_X(j)$ 相等

① 分布已知: H_0: $p_1(j)=p_2(j)=\dots=p_r(j)=p_j^0$ (已知), $j=1,2\dots s$

$K_n=\sum_{i=1}^r\sum_{j=1}^s\frac{(n_{ij}-n_{i.}p_j^0)^2}{n_{i.}p_j^0}$ $D=\{K_n>\chi^2_{(s-1)r,\alpha}\}$ $p(k_0)\doteq P(\chi^2_{(r-1)s}\ge k_0)$

② 分布未知: H_0: $p_1(j)=p_2(j)=\dots=p_r(j)=p_j$, $j=1,2,\dots s$

$K_n=n(\sum_{i=1}^r\sum_{j=1}^s\frac{n_{ij}^2}{n_{i.}n_{.j}}-1)$ $D=\{K_n>\chi^2_{(r-1)(s-1),\alpha}\}$ $p(k_0)\doteq P(\chi^2_{(r-1)(s-1)}\ge k_0)$

HW:

- X from $U(\theta,1)$ LR test: H_0: $\theta\ge\theta_0 \leftrightarrow H_1$: $\theta<\theta_0$

$\lambda(X)=\begin{cases}1, & X_{(1)}\le\theta_0 \\ 0, & X_{(1)}>\theta_0\end{cases}$ $D=\{X_{(1)}<c\}$ $\alpha=\int_{\theta_0}^c\frac{n(1-t)^{n-1}}{(1-\theta_0)^n}dt=1-(\frac{1-c}{1-\theta_0})^n\Rightarrow c=1-(1-\alpha)^{1/n}(1-\theta_0)$

$\Pi(\theta)=\begin{cases}1-\frac{(1-\alpha)(1-\theta_0)^n}{(1-\theta)^n}, & \theta<c \\ 0, & \theta\ge c\end{cases}$

- X from $f(x,\theta)=e^{-(x-\theta)}I_{x>\theta}$, CI for θ

$X_{(1)}$: $f(x)=ne^{n(\theta-x)}I_{x>\theta}$, $Y=X_{(1)}-\theta$, $f(y)=ne^{-ny}I_{y>0}$

$P(a\le X_{(1)}-\theta\le b)=e^{-na}-e^{-nb}=1-\alpha\Rightarrow a=0, b=-\frac{\ln\alpha}{n}$

- calculate power / power function (在H下，分类 H_0 & H_1)

单边 U: $1-\Phi(|u|)$ 双边 $1-\Phi(|u|)+\Phi(-|u|)$ 形式上一致

Properties:

- $X_k\sim N(a_k,\sigma_k^2)$ $T=\sum c_kX_k\Rightarrow T\sim N(\sum a_kc_k,\sum c_k^2\sigma_k^2)$
- $X_i\sim N(a,\sigma^2)$ $Y=AX\Rightarrow Y\sim N(A\vec{\mu},A\vec{\sigma}^2A^T)$
- $X_i\sim N(a,\sigma^2)\Rightarrow\bar{X}\sim N(a,\sigma^2/n)$, $(n-1)S^2/\sigma^2\sim\chi^2_{n-1}$
- $P(\lambda)$: $EX=\lambda$, $\text{var}X=\lambda$, $P(X=k)=\frac{\lambda^k}{k!}e^{-\lambda}$ PGF: $e^{\lambda(s-1)}$

$\sum P(\lambda_i)=P(\sum\lambda_i)$

- $G(p)$: $P(X=k)=(1-p)^{k-1}p$ $EX=1/p$ $\text{var}X=q/p^2$ PGF: $sp/(1-sq)$

n个 $G(p)$ 相加 $NBin(n,p)$

- $\mathcal{E}(\lambda)$: $f(x)=\lambda e^{-\lambda x}I_{x>0}$, $EX=\frac{1}{\lambda}$ $\text{var}X=\frac{1}{\lambda^2}$ MGF: $\frac{\lambda}{\lambda-s}$

n个 $\mathcal{E}(\lambda)$ 相加 → $Gamma(n,\lambda)$

- $X\sim f(x)$ $X_{(1)}$: $g(x)=nf(x)[1-F(x)]^{n-1}$ $X_{(n)}$: $g(x)=nF(x)^{n-1}f(x)$
- $\Gamma(p,\alpha)$ $f(x)=\frac{\alpha^p}{\Gamma(p)}x^{p-1}e^{-\alpha x}I_{x>0}$ $EX=p/\alpha$ $\text{var}X=p/\alpha^2$
- χ_r^2 $f(x)=\frac{1}{2^{r/2}\Gamma(\frac{r}{2})}x^{\frac{r}{2}-1}e^{-\frac{x}{2}}$ $EX=r$ $E\frac{1}{X}=\frac{1}{r-2}$ $\text{var}X=2n$

C.F $(1-2it)^{-r/2}$

- $T_r=\frac{Z}{\sqrt{\chi_r^2/r}}$ $f(x)=\frac{\Gamma(\frac{r+1}{2})}{\Gamma(\frac{r}{2})\sqrt{\pi r}}(1+\frac{t^2}{r})^{-\frac{r+1}{2}}$ $ET=0$ $\text{var}T=r/r-2$

eg	1	2	3	4	$y_{i.}$
1	3	7	5	4	19/4
2	–1	2	0	2	3/4
3	1	2	4	0	7/4
$y_{.j}$	1	11/3	3	2	$y_{..}=\frac{29}{12}$

$\hat\alpha_1=\frac{19}{4}-\frac{29}{12}$

$\hat\beta_1=1-\frac{29}{12}$

$SS_A=4\times[(\frac{7}{4})^2+(\frac{3}{4})^2+(\frac{7}{4})^2]-12\times(\frac{29}{12})^2$ $SS_T=[3^2+7^2+\dots+0^2]-12\times(\frac{29}{12})^2$

→ $SS_e=SS_T-SS_A-SS_B$ $F_A=\frac{SA/2}{SSe/6}$ $F_{2,6,\alpha}$